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Lung Nodule Prognostics

– DS - Intelligent Modeling report –

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Abstract

Text of abstract. Short info about:

- project relevance/importance,
- intelligent methods used for solving,
- data involved in the numerical experiments;
- conclude by the the results obtained.
- Please add a graphical abstract of your work.

Remind that a good report should:

- be fun to read with many figures and visualizations;
- be easy to follow even for AI/ML novice;
- clearly convey the potential of AI/ML to the application domain;
- around 10 minutes to read (although this is not a hard constraint).

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Chapter 1

Introduction

1.1 What? Why? How?

Motivate and abstractly describe the problem you are addressing and how you are addressing it.

- What is the (scientific) problem?
- Why is it important?
- What is your basic approach?

A short discussion of how it fits into related work in the area is also desirable. Summarize the basic results and conclusions that you will present.

1.2 Paper structure and original contribution(s)

The research presented in this paper advances the theory, design, and implementation of several particular models.

The main contribution of this report is to present an intelligent algorithm for solving the problem of ...

The second contribution of this report consists of building an intuitive, easy-to-use and user friendly software application. Our aim is to build an algorithm that will help ...

The third contribution of this thesis consists of ...

The present work contains *xyz* bibliographical references and is structured in five chapters as follows.

The first chapter/section is a short introduction in ...

The second chapter/section describes ...

The chapter/section 4 details . . .

Chapter 2

Scientific Problem

2.1 Problem description

Low-dose computed tomography (CT) screening has drastically improved the early detection of lung cancer, but at the same time, it introduced a clinical dilemma: the management of indeterminate lung nodules. When a nodule is detected, its clinical trajectory is highly uncertain. It may remain harmless and slow growing for a patient’s entire life, but it could also progress into aggressive metastatic lung cancer at a fast pace.

Currently, radiologists and oncologists rely on a subjective synthesis of heterogeneous attributes such as the nodule’s radiological subtype (solid, part-solid, ground-glass), volumetric growth patterns over time, and subtle morphological textures to stratify risk. Because this manual assessment is highly variable among practitioners, it frequently leads to two adverse outcomes: overtreatment, where patients with benign nodules are subject to invasive, risky, and expensive biopsies or undertreatment, where aggressive tumors are not removed in time.

2.2 Why an Intelligent Algorithm is Mandatory

This problem requires an intelligent algorithm because the underlying data exceeds human cognitive and visual processing capacities. The human visual system cannot reliably extract tiny, complex patterns from 3D medical images. Furthermore, rule based clinical guidelines or standard linear statistical models fail because lung cancer progression depends on many complex factors.

Deep learning algorithms are uniquely capable of automatically extracting latent hierarchical representations from raw 3D pixel/voxel data. More importantly, an intelligent, multi-task algorithm is required to computationally map the biological reality that a nodule’s morphology, growth rate, and

ultimate malignancy are deeply interdependent phenomena, not isolated variables.

2.3 Solution Methods

To effectively predict nodule malignancy (T1), progression (T2), and subtype (T3) from medical images, we evaluate two different frameworks. Here, we contrast an approach that treats each task separately with a joint-learning method designed to capture their biological overlap.

2.3.1 Traditional Baseline: Single-Task Learning (STL)

The traditional baseline approach utilizes Single-Task Learning, where separate networks are trained independently for malignancy, progression, and subtype. The primary advantage of these models is that they are straightforward to implement and optimize. Hyperparameter tuning is relatively simple, and the loss landscape for a single objective is generally well behaved. However, a major disadvantage is that STL completely ignores the biological correlations between the tasks. It forces the models to learn redundant feature extractors from scratch for each task. This is highly inefficient and increases the risk of overfitting, which is a critical flaw when working with limited medical datasets like LIDC-IDRI or LNDb.

2.3.2 Proposed Solution: Multi-Task Learning (MTL) with Cross-Talk

The proposed solution employs Multi-Task Learning equipped with cross-talk. By jointly predicting T1, T2, and T3, MTL acts as a strong inductive bias and a natural regularizer. Shared layers allow the model to leverage larger, combined pools of information, making the learned representations far more robust. The advantage of cross-talk mechanisms, such as cross-stitch units or co-attention, is that they elevate this by dynamically controlling how information flows between task specific branches. This allows the model to autonomously determine, for example, how much a specific visual texture should influence the overall malignancy prediction. On the other hand, a significant disadvantage is that MTL architectures are difficult to tune. They are highly susceptible to negative transfer, where optimizing one task degrades the performance of another, or task domination, where the loss of a complex task overwhelms a simpler one.

2.4 Formal Problem Definition

The objective is to formulate a multi-task learning mapping function capable of joint inference.

2.4.1 Inputs

Let the input space be defined as $X = \{\dots\}$, where:

...

2.4.2 Outputs

The MTL model learns a parameterized function $f_\theta : X \rightarrow (\hat{Y}_1, \hat{Y}_2, \hat{Y}_3)$ to predict three distinct target spaces:

- T1 Nodule malignancy risk: $\hat{Y}_1 \in \{0, 1\}$, where 0 is Benign and 1 is Malignant.
- T2 2 year progression probability: $\hat{Y}_2 \in \{0, 1\}$, where 0 is Stable and 1 is Progressive.
- T3 Radiological subtype classification: $\hat{Y}_3 \in \{0, 1, 2\}$, representing a multiclass prediction (0 = solid, 1 = part-solid, 2 = ground-glass).

2.4.3 Objective

The network optimizes task specific parameters $\theta_1, \theta_2, \theta_3$, potentially alongside a foundational shared backbone θ_{shared} , and cross-talk routing parameters α (cross-stitch matrices / co-attention weights).

These parameters are jointly trained to minimize a globally weighted loss function:

$$\mathcal{L}_{total} = \lambda_1 \mathcal{L}_{BCE}(\hat{Y}_1, Y_1) + \lambda_2 \mathcal{L}_{BCE}(\hat{Y}_2, Y_2) + \lambda_3 \mathcal{L}_{CE}(\hat{Y}_3, Y_3)$$

where $\mathcal{L}_{BCE}$ is Binary Cross-Entropy, $\mathcal{L}_{CE}$ is Categorical Cross-Entropy, and λ_i are dynamic weighting hyperparameters to balance task gradients.

2.5 Motivation

This problem directly impacts patient survival. An accurate, generalized model can fundamentally alter the clinical workflow for pulmonary oncology. By accurately identifying high risk nodules early, life saving treatments can start sooner. On the flip side, by confidently identifying indolent nodules, the system reduces the immense psychological and financial cost of unnecessary biopsies and repeated radiation exposure.

From an AI perspective, translating deeply interconnected biological traits into a computational architecture is a state of the art challenge. Designing cross-talk mechanisms that avoid negative transfer in noisy, high dimensional medical data pushes the boundaries of network design. Furthermore, by including SHAP explanations, we transform this project from a purely mathematical exercise into a trustworthy clinical tool. It tackles the black box problem in medical AI, ensuring that doctors can actually see why the algorithm made its prediction and trust it to help save lives.

Chapter 3

Related work

Recent advances in deep learning have significantly improved automated analysis of medical imaging data, particularly in lung cancer screening using computed tomography (CT). Lung nodules detected during low-dose CT screening present a major diagnostic challenge, as their clinical significance varies widely: some nodules remain stable for years, while others rapidly progress to malignancy. Accurate risk stratification therefore requires the integration of multiple heterogeneous sources of information, including imaging characteristics, volumetric growth patterns, and patient clinical data.

Multi-task learning (MTL) has been applied to improve lung nodule classification in CT images. The authors of [9] propose a multi-task CNN (MT-CNN) that distinguishes malignant from benign nodules using two tasks: a main classification task and an auxiliary image reconstruction task. The model extracts nine 2D views from each 3D nodule and processes each view with a shared network followed by two branches: one for classification and one for reconstruction. The final prediction is obtained by a weighted fusion of the nine view-specific models, allowing the network to capture 3D spatial information while maintaining lower computational complexity.

The method is evaluated on two publicly available datasets, LIDC-IDRI and LUNA-16. LIDC-IDRI contains 1,018 annotated CT scans with radiologist-provided malignancy scores, while LUNA-16 is a subset of LIDC-IDRI designed for nodule detection and classification tasks.

The proposed MT-CNN achieves strong performance compared to state-of-the-art approaches, reaching an AUC of 97.3% on LUNA-16 and 95.59% on LIDC-IDRI, along with a low false positive rate of 3.2%. These results demonstrate that incorporating an auxiliary reconstruction task helps preserve fine-grained image features and improves classification accuracy.

Our work differs in that we address a prognostic setting with multiple clinically meaningful tasks: malignancy risk prediction, two-year progression probability, and radiological subtype classification. Instead of using reconstruction as an auxiliary task, our model jointly learns these related clinical

endpoints. Our approach is more informative for clinical decision support because it predicts several complementary outcomes simultaneously and analyzes their interactions, providing richer insights than a single malignancy classification task.

MTL has also been explored to improve both prediction accuracy and interpretability in lung nodule analysis. Wang et al.[8] propose IMAL-Net, an interpretable multi-task attention network for screening invasive lung adenocarcinoma from CT images. Their method jointly performs two tasks: lung nodule segmentation and invasiveness classification, using two sub-ResNet networks connected through a prior-attention mechanism. The segmentation branch provides spatial priors that guide the classification branch, while radiomic features extracted from segmentation masks are fused with deep semantic features to improve both performance and interpretability.

The model is trained and evaluated on a large, non-public clinical dataset consisting of 1,626 patients collected from two hospitals, with biopsy-confirmed labels, and an additional external validation set of 200 nodules. This makes the dataset clinically reliable but not publicly available.

IMAL-Net achieves strong performance, reaching an AUC of 93.8% and a recall of 93.8% in identifying invasive lung adenocarcinoma. The results show that combining segmentation, attention mechanisms, and radiomic-semantic feature fusion improves both classification accuracy and model interpretability.

In our work, we focus on lung nodule prognostics with three prediction tasks: malignancy risk, two-year progression probability, and radiological subtype classification. Instead of combining segmentation and classification only, our model learns multiple clinically meaningful endpoints simultaneously and explicitly models interactions between them. Our approach provides richer clinical information because it predicts several complementary outcomes within a single framework. By jointly modeling malignancy, progression risk, and radiological subtype, the model can capture relationships between these factors and support more comprehensive clinical decision making.

Multi-task learning has also been used to jointly exploit the relationships between different lung nodule analysis tasks. Wang et al.[7] propose DeepLN, a multi-task deep learning framework that simultaneously predicts imaging characteristics, malignancy, and pathological subtypes from CT scans. The model is based on a 3D-ResNet backbone and incorporates auxiliary tasks, such as nodule density and morphological feature classification, to guide the main tasks and improve feature representation.

The model is trained on a large retrospective clinical dataset of 8,950 pulmonary nodules from 5,823 patients, with pathology-confirmed labels serving as the gold standard. This dataset is not publicly available but provides strong clinical reliability.

DeepLN demonstrates strong performance across multiple tasks. For malignancy prediction, it achieves an AUC of 0.8503 on the test set, while for pathological subtype classification, it reaches AUCs of 0.8675 for adenocarcinoma and 0.8792 for squamous cell carcinoma. Additionally, the inclusion of auxiliary imaging features improves performance compared to single-task models, highlighting the benefit of multi-task learning.

In contrast, our work focuses on lung nodule prognostics rather than combining imaging feature prediction with classification tasks. Our model jointly predicts malignancy risk, two-year progression probability, and radiological subtype, which represent clinically meaningful endpoints rather than intermediate imaging features. By modeling these related outcomes together, our approach provides more comprehensive clinical decision support and captures relationships between biological and radiological factors.

Another multi-task approach is proposed by Ni et al.[4], who introduce a two-stage multitask U-Net framework for pulmonary nodule segmentation and malignancy risk prediction. Their method follows a coarse-to-fine strategy, where a 3D multiscale U-Net (MU-Net) first localizes the nodule region, followed by a 2.5D multiscale separable U-Net (MSU-Net) that refines segmentation and performs multi-task classification. The model jointly predicts malignancy along with radiological characteristics such as margin, texture, and calcification by sharing features between segmentation and classification branches.

The approach is evaluated on the public LIDC-IDRI dataset, using 1,561 annotated nodules with radiologist-provided labels. Experimental results show strong performance, achieving a Dice similarity coefficient of 83.04% for segmentation and an AUC of 84.3% for malignancy prediction. The results demonstrate that integrating segmentation and classification in a multitask framework improves both feature representation and predictive accuracy.

While this approach improves diagnostic performance by leveraging segmentation cues and radiological feature prediction, it focuses primarily on detection and characterization tasks. In contrast, our work addresses a broader prognostic setting by jointly modeling malignancy risk, progression probability, and radiological subtype. This formulation enables the model to capture dependencies between clinically meaningful outcomes and provide richer insights into lung nodule behavior.

Radiomics-based approaches have also been proposed to predict the growth behavior of pulmonary nodules. Sun et al.[5] develop a CT-based radiomics model to predict the growth or long-term stability of pulmonary ground-glass nodules (GGNs). Their method involves manual segmentation of nodules followed by extraction of high-dimensional radiomic features (1218 features), including intensity, shape,

and texture descriptors. Feature selection is performed using mRMR and LASSO, and a radiomics signature is constructed via logistic regression.

The model is trained and validated on a dataset of 253 patients with 1115 CT images, split into training (70%) and validation (30%) sets. A combined nomogram integrating radiomics features with clinical factors (age, size, and location) achieves the best predictive performance, with AUC values of 0.843 in the training set and 0.824 in the validation set. The radiomics model significantly outperforms traditional clinical features, demonstrating the advantage of capturing tumor heterogeneity through quantitative imaging descriptors.

However, this approach focuses on a single prediction task, namely distinguishing between growing and stable nodules. In contrast, our work adopts a multi-task learning framework that simultaneously predicts malignancy risk, progression probability, and radiological subtype. By jointly modeling multiple clinically relevant outcomes, our approach captures interdependencies between different aspects of lung nodule behavior and provides more comprehensive prognostic information for clinical decision making.

Another related approach focuses on the automatic categorization of pulmonary nodules based on their radiological characteristics. Tu et al.[6] propose a convolutional neural network (CNN)-based computer-aided diagnosis (CADx) system for the automatic categorization of solid, part-solid, and non-solid pulmonary nodules in CT images. Their method operates directly on two-dimensional regions of interest (ROIs) without requiring explicit nodule segmentation, allowing the model to learn hierarchical features directly from image context and avoid errors associated with segmentation.

The framework supports both classification (three nodule types) and regression (radiological scoring from 1 to 5), and is trained on the LIDC dataset containing annotations from multiple radiologists. Experimental results show that the CNN significantly outperforms traditional histogram-based methods and achieves substantial agreement with expert radiologists, with Cohen's kappa values around 0.74.

While this work focuses on improving automated categorization and scoring of nodule types, it primarily addresses a single aspect of nodule characterization. In contrast, our approach extends this idea by incorporating radiological subtype classification as one of several tasks within a multi-task learning framework. By jointly predicting radiological subtype together with malignancy risk and progression probability, our model leverages shared representations across tasks and provides a more comprehensive characterization of lung nodules for clinical decision support.

A different multi-task strategy for lung nodule analysis is proposed in MTMR-Net [3]. This model

jointly performs benign–malignant classification and regression of multiple radiological semantic attributes, including spiculation, lobulation, sphericity, calcification, and others. The architecture is based on a shared ResNet backbone with two task-specific branches: a classification module and a regression module, where attribute features are explicitly fed into the classification branch to model a cause-and-effect relationship between semantic characteristics and malignancy. In addition, the model incorporates a Siamese network trained with a margin ranking loss to improve discrimination between ambiguous nodules by enforcing correct ordering of malignancy scores.

The method is evaluated on the publicly available LIDC-IDRI dataset, which contains 1,018 CT scans and 1,422 annotated nodules with malignancy scores and radiological attributes provided by multiple radiologists. The experiments are conducted using five-fold cross-validation to ensure robustness.

Experimental results show that MTMR-Net achieves strong performance, reaching an accuracy of 93.5%, sensitivity of 93.0%, and AUC of 0.979 for malignancy classification, outperforming several state-of-the-art methods. Additionally, the model achieves lower absolute error in attribute score regression compared to traditional methods such as LASSO and elastic net, demonstrating the effectiveness of jointly modeling classification and semantic attributes.

In contrast, our work focuses on lung nodule prognostics with multiple clinical endpoints rather than malignancy classification alone. Besides predicting malignancy risk, our model also estimates two-year progression probability and radiological subtype, capturing complementary aspects of nodule behavior over time. This approach provides richer clinical insight by modeling interactions between several prognostic outcomes and better reflects the biological complexity of lung nodules for clinical decision making.

Recent work has also explored the integration of domain-specific semantic information into deep learning models for lung nodule analysis. Dong et al. [2] propose a 3D convolutional neural network (SCCNN) that incorporates radiological semantic characteristics as auxiliary tasks within a multi-task learning framework. The model extracts multi-view 3D patches of lung nodules and processes them using a shared CNN backbone with task-specific branches for malignancy classification and semantic feature prediction. In addition, a Convolutional Block Attention Module (CBAM) is introduced to enhance feature fusion by focusing on the most informative spatial and channel-wise features.

The model is trained and evaluated on the publicly available LIDC-IDRI dataset, which contains over 1,000 CT scans with annotations from multiple radiologists, including semantic characteristics such as lobulation, spiculation, subtlety, and malignancy scores. After preprocessing and data cleaning,

1,779 nodules are used for training and evaluation.

Experimental results show that the proposed SCCNN significantly outperforms a baseline 3D CNN. The model achieves an AUC of up to 0.973 and an accuracy of 95.45% when combining semantic characteristics and attention mechanisms, compared to 89.77% accuracy for the baseline model. These results demonstrate that incorporating semantic features improves both classification performance and interpretability.

In contrast, our work focuses on lung nodule prognostics with multiple clinical endpoints. Instead of predicting malignancy alone, our model jointly estimates malignancy risk, two-year progression probability, and radiological subtype. This allows us to capture relationships between different aspects of nodule behavior and provide more comprehensive prognostic information for clinical decision making.

In addition to task-specific approaches, several studies have reviewed the broader landscape of multi-task learning in deep neural networks. Crawshaw[1] provides a comprehensive survey of multi-task learning methods and their applications across different domains. The survey discusses the main categories of MTL architectures, including hard parameter sharing, soft parameter sharing, cross-stitch networks, and attention-based task interaction mechanisms. It also highlights the benefits of MTL in improving generalization, reducing overfitting, and enabling knowledge transfer between related tasks. The survey emphasizes that MTL is particularly effective when tasks share underlying representations, which has motivated its increasing adoption in complex prediction problems such as medical image analysis.

Chapter 4

Investigated approach

4.1 Proposed approach - methodology

Describe your approach!

- Exploratory data Analysis
- Training and testing the model
- Validation and comparisons

Describe in reasonable detail the algorithm you are using to address this problem. A psuedocode description of the algorithm you are using is frequently useful. Trace through a concrete example, showing how your algorithm processes this example. The example should be complex enough to illustrate all of the important aspects of the problem but simple enough to be easily understood. If possible, an intuitively meaningful example is better than one with meaningless symbols.

4.2 Numerical validation

Explain the experimental methodology and the numerical results obtained with your approach and the state of art approache(s).

Try to perform a comparison of several approaches.

Statistical validation of the results.

4.2.1 Methodology

- What are criteria you are using to evaluate your method?

- What specific hypotheses does your experiment test? Describe the experimental methodology that you used.
- What are the dependent and independent variables?
- What is the training/test data that was used, and why is it realistic or interesting? Exactly what performance data did you collect and how are you presenting and analyzing it? Comparisons to competing methods that address the same problem are particularly useful.

4.2.2 Data

The primary dataset utilized in this study is the LIDC-IDRI (Lung Image Database Consortium and Image Database Resource Initiative) dataset. As a premier public resource for lung nodule analysis, it provides a comprehensive collection of thoracic CT scans, unblinded radiologist annotations, and clinical diagnostic records. Because nodules are frequently reviewed by multiple radiologists, the dataset captures inherent clinical variability and uncertainty, making it highly representative of real-world diagnostic scenarios. For this analysis, the data is divided into three interconnected components: the raw imaging data, the clinical diagnosis records and the nodule annotations.

4.2.2.1 Radiological Imaging Data

The imaging data consists of 3D CT scans stored as raw DICOM files. The pipeline recursively extracts files with the "CT" modality and reconstructs the full axial chest volume by sorting the 2D slices along the Z-axis (utilizing spatial attributes such as ImagePositionPatient). Each 2D slice has a standard image size of 512x512 pixels. To ensure accurate radiodensity representations, the raw pixel arrays are converted into standardized Hounsfield Units (HU) using the RescaleSlope and RescaleIntercept metadata.

4.2.2.2 Radiologist Nodule Annotations

Accompanying the DICOM files are XML annotation files documenting findings from radiologist reading sessions. These annotations describe the location, size, shape, internal structure, margin sharpness, and visual appearance of the identified lesions. A critical characteristic extracted for this study is the nodule texture, which indicates how dense or solid a lesion appears on a CT scan.

The original texture scores in the dataset range from 1 to 5, each corresponding to the following labels, in order: non-solid, mostly non-solid, part-solid, mostly solid, solid

To reduce noise and facilitate comparative analysis, these scores were mapped into three distinct radiological categories:

Ground-Glass (Scores 1 & 2): Lighter, less dense regions.

Part-Solid (Score 3): Lesions containing both dense and non-dense regions.

Solid (Scores 4 & 5): Denser, brighter regions.

4.2.2.3 Clinical Diagnosis Data

A supplementary clinical dataset provides patient- and nodule-level diagnostic context across 157 patient records. This component contains variables detailing clinical outcomes, including the overall diagnosis result, tumor location, and the specific method used to confirm the diagnosis (e.g., biopsy, surgical resection, radiological follow-up, or progression). For metastatic cases, the primary tumor site is recorded across 112 unique classes (e.g., non-small cell lung cancer, melanoma, colon cancer).

4.2.3 Data Processing and Feature Engineering

To structure the data for modeling, the annotation data was aggregated into two distinct relational tables.

One is a detailed table with one row for each texture annotation, containing the following information: Patient ID, Nodule ID, Reading session index, Texture score and Texture category.

The other table is a patient-level summary table that combines all annotations for a patient and stores the following data: Total number of texture annotations, Number of ground-glass nodules, Number of part-solid nodules, Number of solid nodules, Dominant texture category.

The dominant texture category is the category that appears most often for that patient. For example, if most nodules for a patient are solid, then the patient is labeled as having a dominant solid texture pattern.

4.2.4 Machine Learning Target Variables

Predicting exact medical diagnoses across highly granular classes is challenging, particularly given the sample size constraints of the clinical dataset. To create robust predictive models, the clinical labels were simplified into three distinct binary and multiclass targets, focusing on the most critical medical distinctions:

Malignancy prediction: 0 for benign (derived from "benign/non-malignant" labels) and 1 for malignant (derived from "malignant primary lung cancer" and "malignant metastatic lesion" labels);

Progression prediction: 0 for stable (nodule remained stable for at least two years) and 1 for progressive (documented progression);

Radiological subtype: as previously mentioned: 0 for solid, 1 for part-solid and 2 for ground-glass

This makes the classification problem easier and helps the model focus on the most important medical distinction: whether the nodule is dangerous or not.

Overall, the dataset is valuable because it combines CT images, radiologist annotations, and diagnosis information. This supports both exploratory analysis and predictive modeling. Furthermore, the presence of missing values, annotation variability, and multiple expert opinions makes the dataset more representative of real clinical scenarios.

4.2.5 EDA

After extracting and inspecting the data, several properties relevant to modeling were identified.

Regarding texture distribution, solid nodules are by far the most frequent category, followed by ground-glass and part-solid. This class imbalance is consistent with clinical literature and must be accounted for during training. Furthermore, most CT series contain only one or two annotated nodules. The distribution of nodule counts per scan is heavily right-skewed, meaning a small number of scans account for a disproportionately large share of the total annotations.

Because each patient may have been reviewed by up to four radiologists in separate reading sessions, the same nodule can receive different texture scores from different readers. This inter-reader variability was partially mitigated at the patient level by applying a majority-vote approach to establish a single "dominant texture category" across all annotations.

In the diagnosis dataset, the nodule-level columns for higher-indexed nodules (e.g., nodules 4 and 5) contain a massive fraction of missing values, reflecting the fact that most patients have fewer than four annotated nodules. Additionally, the *primary_tumor_site* column was excluded as a feature; it applies specifically to metastatic cases and suffers from extreme high cardinality (containing over 110 unique free-text classes for just 157 patients), rendering it too fragmented for robust model training.

Finally, both derived classification targets exhibit class imbalance. The malignancy target is moderately skewed toward the benign class, while the progression target is severely unbalanced, with stable cases substantially outnumbering progressive ones.

4.2.6 Results

Present the quantitative results of your experiments. Graphical data presentation such as graphs and histograms are frequently better than tables. What are the basic differences revealed in the data. Are

they statistically significant?

4.3 Discussion

- Is your hypothesis supported?
- What conclusions do the results support about the strengths and weaknesses of your method compared to other methods?
- How can the results be explained in terms of the underlying properties of the algorithm and/or the data.

Chapter 5

Conclusion and future work

Try to emphasise the strengths and the weaknesses of your approach. What are the major shortcomings of your current method? For each shortcoming, propose additions or enhancements that would help overcome it. Add all four components of a SWOT analysis.

Briefly summarize the important results and conclusions presented in the paper.

- What are the most important points illustrated by your work?
- How will your results improve future research and applications in the area?

Chapter 6

Latex examples

Item example:

- content of item1
- content of item2
- content of item3

Figure example

... (see Figure 6.1)

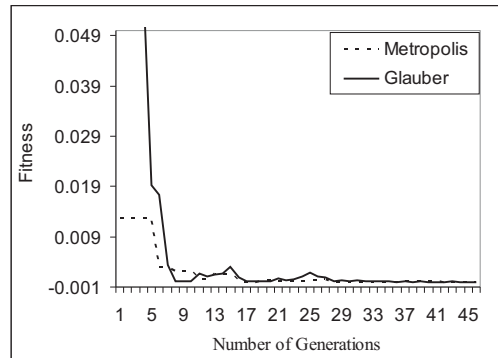


Figure 6.1: The evolution of the swarm size during the GA generations. This results were obtained for the f_2 test function with 5 dimensions.

Table example: (see Table 6.1)

Algorithm example

... (see Algorithm 1).

Table 6.1: The parameters of the PSO algorithm (the micro level algorithm) used to compute the fitness of a GA chromosome.

Parameter	Value
Number of generations	50
Number of function evaluations/generation	10
Number of dimensions of the function to be optimized	5
Learning factor c_1	2
Learning factor c_2	1.8
Inertia weight	$0.5 + \frac{rand()}{2}$

Algorithm 1 SGA - Spin based Genetic AAlgorithm

BEGIN

@ Randomly create the initial GA population.

@ Compute the fitness of each individual.

for i=1 TO NoOfGenerations **do**

for j=1 TO PopulationSize **do**

 p $\leftarrow$ RandomlySelectParticleFromGrid();

 n $\leftarrow$ RandomlySelectParticleFromNeighbors(p);

 @ Crossover(p, n, off);

 @ Compute energy ΔH

if ΔH satisfy the Ising condition **then**

 @ Replace(p,off);

end if

end for

end for

END

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